How Robust Is the Effect of Data Analytics on SME Performance? A Specification Curve Analysis of Bar-Gill, Brynjolfsson, and Hak (2024)

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Abstract

Bar-Gill, Brynjolfsson, and Hak (2024) report that access to eBay's Seller Hub analytics dashboard increases weekly seller revenue by 3.6% ($\hat{\delta} = 0.035$, $p < 0.01$) using a log(GMV+1) dependent variable with seller and week fixed effects. We conduct a specification curve analysis (Simonsohn, Simmons, and Nelson, 2020) to assess the robustness of this intent-to-treat estimate across 160 specifications that systematically vary four modeling dimensions: functional form of the dependent variable (7 forms), fixed effects structure (4 structures), sample definition (3 samples), and covariate design (2 designs). Using a 10% random subsample of the original data, we find that the treatment effect estimate is qualitatively sensitive to the fixed effects structure, which accounts for 61.8% of the explained variation in a meta-regression across specifications. Dropping week fixed effects reverses the sign of the estimated effect, indicating that aggregate platform trends are a dominant confounder in this staggered design. Functional form, the focus of the Chen and Roth (2024) critique, contributes only 1.2% of specification-level variation. These findings highlight the importance of transparent sensitivity analysis in experimental program evaluations.

JEL Codes: C21, C93, L81, M15
Keywords: specification curve, sensitivity analysis, data analytics, SME performance, fixed effects

1 Introduction

A growing literature studies the causal effects of data analytics tools on firm performance. Bar-Gill, Brynjolfsson, and Hak (2024) contribute to this literature by evaluating eBay's Seller Hub (SH), an analytics dashboard that provides small and medium enterprise (SME) sellers with detailed performance metrics, trend visualizations, and listing optimization tools. Exploiting a staggered randomized controlled trial in which 184,222 sellers were invited to SH across seven rollout cohorts

between May and August 2016, the authors estimate an intent-to-treat (ITT) effect of $\hat{\delta} = 0.035$ (SE $= 0.005$) on log weekly gross merchandise volume (GMV), corresponding to an approximately 3.6% increase in sales.

This estimate is obtained from a generalized difference-in-differences (DiD) specification with seller and week fixed effects, using $\log(\text{GMV} + 1)$ as the dependent variable. While the experimental design provides a credible source of identification, the reported estimate depends on several modeling choices that are standard in applied work but potentially consequential: the functional form used to handle zero-valued observations, the fixed effects structure, the sample definition, and the covariate design. The original paper does not systematically investigate how these choices affect the main finding.

We apply the specification curve analysis framework of Simonsohn, Simmons, and Nelson (2020) to assess the sensitivity of the ITT estimate to these modeling decisions. Our analysis varies four dimensions: (i) the functional form of the dependent variable (7 forms), motivated by the Chen and Roth (2024) critique of the $\log(1+y)$ transformation; (ii) the fixed effects structure (4 structures), motivated by the staggered design's dependence on absorbing aggregate time trends (Goodman-Bacon, 2021); (iii) the sample definition (3 samples); and (iv) the covariate design (2 designs). This yields 160 valid specifications, which we summarize using a specification curve plot and a meta-regression that decomposes cross-specification variation into contributions from each dimension.

Our main finding is that the fixed effects structure is the most consequential modeling decision. Dropping week fixed effects, which absorb aggregate platform trends, shifts the average estimate by -8.83 units in the meta-regression ($p = 0.046$) and flips the sign of the treatment effect in many specifications. This pattern is consistent with concerns about staggered DiD designs raised by Goodman-Bacon (2021): in the absence of week fixed effects, the estimate confounds the treatment effect with differential exposure to eBay-wide trends across rollout cohorts. Covariate design is the second most important dimension (29.1% of meta-regression R^2), while functional form, despite being the focus of important methodological work by Chen and Roth (2024), contributes only 1.2%.

The remainder of this paper is organized as follows. Section 2 describes the data. Section 3 reviews the authors' econometric specification. Section 4 details our specification curve design. Section 5 presents the results. Section 6 concludes.

2 Data

The data come from the replication package of Bar-Gill, Brynjolfsson, and Hak (2024). The original study uses a panel of 184,222 eBay sellers observed at weekly frequency during the Seller Hub rollout period. We use a 10% random subsample containing 18,422 sellers and 423,706 seller-week observations. Because the full data are proprietary, we focus on qualitative patterns rather than point-estimate magnitudes; the original paper's $\hat{\delta} = 0.035$ serves as the benchmark.

Table 1 reports summary statistics for the analysis sample. Mean weekly GMV is \$930.70 with substantial right skew (median \$225.00, maximum \$690,212). The distribution features a mass point at zero: 24.5% of seller-weeks record no sales, a feature that motivates the functional-form dimension of our specification curve. Mean quantity sold is 31.14 units per week, with a similar zero-mass structure. The SH access rate across all seller-weeks in the sample is 23%, reflecting the staggered rollout; the adoption rate (sellers who actively used SH) is 7%, indicating substantial non-compliance that motivates the ITT estimand.

3 Model Specification

The authors' baseline specification (Equation 2, p. 7352) is a generalized DiD model:

$$\log(\text{Sales}_{sw} + 1) = \alpha_s + \beta_w + \delta \cdot \text{SHAccess}_{sw} + \varepsilon_{sw} \tag{1}$$

where s indexes sellers, w indexes weeks, α_s are seller fixed effects, β_w are week fixed effects, SHAccess_{sw} is a binary indicator equal to one after seller s's rollout cohort is invited to Seller Hub,

Table 1: Summary Statistics

Variable	N	Mean	Std. Dev.	Median	Min	Max	% Zero
Weekly GMV ($)	423,706	930.70	4,056.35	225.00	0.00	690,212	24.5
Quantity Sold	423,706	31.14	162.02	5.00	0	12,086	24.5
Price ($)	319,925	324.64	1,031.96	136.14	0.01	138,000	—
New Listings	423,706	20.97	165.51	2.00	0	20,858	43.0
Feedback Score	423,660	4,817	16,879	1,172	0	675,599	—
SH Access	423,706	0.23	0.42	0.00	0	1	—
SH Adoption	423,706	0.07	0.26	0.00	0	1	—

Notes: 10% random subsample of the original panel (18,422 sellers, 423,706 seller-weeks). GMV = gross merchandise volume. SH Access = binary indicator for post-invitation weeks. SH Adoption = binary indicator for weeks after the seller actively used Seller Hub. Price is conditional on positive sales.

and δ is the ITT effect. Standard errors are clustered at the seller level.

Several features of this specification deserve scrutiny. First, the $\log(\text{GMV}+1)$ transformation handles the 24.5% of observations with zero sales, but Chen and Roth (2024) show that the resulting coefficient $\hat{\delta}$ lacks a clean percentage interpretation and depends on the arbitrary choice of the additive constant (here, $1). Second, the two-way fixed effects structure ($\alpha_s + \beta_w$) is standard in panel DiD but plays a particularly important role in this staggered design: the seven rollout cohorts receive treatment at different calendar dates, so week fixed effects absorb platform-level trends that could generate spurious correlation between treatment timing and outcomes. Third, the model includes no covariates beyond fixed effects, relying entirely on randomization for identification.

4 Analysis Design

We construct a specification curve (Simonsohn, Simmons, and Nelson, 2020) by systematically varying four dimensions of the baseline model: functional form of the dependent variable, fixed effects structure, sample definition, and covariate design. Table 2 summarizes the full set of modeling choices. The baseline specification (row marked with *) corresponds to the authors' original model. Each alternative is motivated by the econometrics literature or by features of the experimental design; we discuss the rationale for each dimension below.

Table 2: Specification Curve Design

Dimension	ID	Specification	Rationale
Functional form of dependent variable (7 forms)			
	F1*	$\log(\text{GMV}+1)$	Authors' baseline (Eq. 2, p. 7352)
	F2	$\log(\text{GMV}/1000+1)$	Unit-dependence test (Chen and Roth, 2024)
	F3	$\mathbf{1}(\text{GMV}>0)$	Extensive margin (LPM)
	F4	$\log(\text{GMV}) \mid \text{GMV}>0$	Intensive margin
	F5	$\text{asinh}(\text{GMV})$	IHS transformation (Burbidge, Magee, and Robb, 1988)
	F6	GMV in levels	No transformation; dollar change
	F7	Poisson PPML	Chen and Roth (2024) preferred estimator
Fixed effects structure (4 structures)			
	E1*	Seller + week FE	Authors' baseline
	E2	Seller FE only	Tests role of aggregate time trends
	E3	Week FE only	Tests role of seller heterogeneity
	E4	No FE (pooled OLS)	Diagnostic benchmark
Sample definition (3 samples)			
	S1*	Full sample	Authors' baseline ($N = 18{,}422$)
	S2	Positive-sales only	Excludes zero-GMV weeks
	S3	Early adopters	Authors' subsample (Table 2, Col. 4)
Covariate design (2 designs)			
	D1*	No additional controls	FE + randomization only
	D2	Cohort-specific trends	RampGroup $\times$ linear week
Standard-error method (3 methods)			
	C1*	Seller-clustered	Authors' baseline (p. 7353)
	C2	HC1 (robust)	Heteroskedasticity-robust
	C3	Two-way (seller + week)	Cameron, Gelbach, and Miller (2011)

Notes: $\star$ = authors' baseline. The specification grid is $7 \times 4 \times 3 \times 2 = 168$ nominal specifications. We exclude 8 invalid combinations (F3$\times$S2: extensive margin undefined for positive-sales sample), yielding 160 valid specifications. Each is estimated with all three SE methods, producing 480 coefficient–SE pairs. SE methods affect inference only, not point estimates.

4.1 Functional Form

The authors use $\log(\text{GMV}+1)$ throughout, a common transformation for handling zero-valued outcomes. However, 24.5% of seller-weeks record zero sales, and Chen and Roth (2024) show that $\log(1+y)$ creates three problems in this setting: (a) the coefficient is not invariant to the units of y, (b) the additive constant \$1 is arbitrary relative to a distribution with mean \$931 and median \$225, and (c) the estimate lacks a clean percentage interpretation when zeros are prevalent, instead mixing

extensive-margin (whether a seller sells at all) and intensive-margin (how much, conditional on selling) responses.

We test the full range of alternatives recommended by the literature. F2 rescales the outcome to directly demonstrate unit-dependence. F3 and F4 separately estimate the extensive and intensive margins that $\log(1+y)$ conflates. F5 uses the inverse hyperbolic sine (Burbidge, Magee, and Robb, 1988), which handles zeros but retains partial scale-dependence. F7 implements Poisson PPML (Santos Silva and Tenreyro, 2006), which Chen and Roth (2024) recommend as a scale-invariant alternative with a clean semi-elasticity interpretation. F6 serves as a no-transformation benchmark.

As a supplementary analysis, we conduct a *c-sensitivity* exercise: holding all other choices at baseline, we vary the constant c in $\log(\text{GMV}+c)$ over $c \in \{0.01, 0.1, 1, 10, 100, 1000\}$ to trace out the dependence of $\hat{\delta}$ on this arbitrary parameter.

4.2 Fixed Effects Structure

The two-way FE structure ($\alpha_s + \beta_w$) plays a particularly important role in this staggered design. The seven rollout cohorts receive treatment at different calendar dates between May 12 and August 8, 2016, so treatment timing is mechanically correlated with calendar time. Week FE absorb platform-level trends (demand shocks, seasonal patterns, macroeconomic conditions) that could generate spurious correlation between treatment status and outcomes. Goodman-Bacon (2021) show that TWFE estimators in staggered designs decompose into a weighted average of 2×2 DiD comparisons, some of which can receive negative weights; absorbing common time shocks via β_w is essential to prevent these comparisons from confounding the treatment effect with aggregate trends.

E2 (seller FE only) tests this directly: if week FE are "unnecessary" because randomization ensures common trends cancel, E2 should produce a similar estimate to E1. E3 (week FE only) tests the converse: if randomization is valid, as confirmed by the balance test ($\chi^2 = 8.65$, $p = 0.19$, Section 6.1), seller FE improve efficiency but are unnecessary for consistency. E4 serves as a diagnostic benchmark, quantifying the total bias from omitting both sets of fixed effects.

4.3 Sample Definition and Covariate Design

The sample dimension tests two concerns. First, restricting to positive-sales weeks (S2) removes the zero-mass that drives the $\log(1+y)$ critique, isolating the intensive margin. Second, the early-adopter subsample (S3) captures complier heterogeneity: the original paper notes that IV estimates of the treatment-on-the-treated may be biased when the exclusion restriction is violated (Section 4, p. 7354), so this subsample restricts to high-compliance sellers.

The covariate dimension addresses a gap in the original analysis. While the paper tests for parallel pre-trends (Section 6.1, Table E.3), all pre-trend interaction coefficients are estimated relative to the August 8 cohort and are insignificant. However, the main specification never directly controls for cohort-specific trends. D2 adds rollout-group $\times$ linear-week interactions, allowing each of the seven cohorts to have a differential time trajectory. If the treatment effect attenuates under D2, it suggests that residual trend heterogeneity confounds the baseline estimate.

4.4 Meta-Regression

To decompose the sources of cross-specification variation, we estimate a meta-regression across the 160 specifications:

$$\hat{\delta}_k = \gamma_0 + \gamma_F' \mathbf{FuncForm}_k + \gamma_E' \mathbf{FE}_k + \gamma_S' \mathbf{Sample}_k + \gamma_D \cdot \text{CovDesign}_k + u_k \tag{2}$$

where $\hat{\delta}_k$ is the point estimate from specification k, $\mathbf{FuncForm}_k$ is a vector of 6 indicators for F2–F7 (F1 omitted), $\mathbf{FE}_k$ contains 3 indicators for E2–E4 (E1 omitted), $\mathbf{Sample}_k$ contains 2 indicators for S2–S3 (S1 omitted), and CovDesign_k is an indicator for D2. The intercept γ_0 represents the predicted estimate under the baseline specification. A partial R^2 decomposition reveals each dimension's contribution to the total explained variation in $\hat{\delta}$.

5 Analysis Results

5.1 Specification Curve

Figure 1 presents the specification curve. Each of the 160 specifications is plotted as a point, sorted by the magnitude of $\hat{\delta}$. The lower panel indicates which modeling choices define each specification. Approximately 55% of specifications produce negative point estimates, and the sign of the treatment effect depends on the combination of modeling choices. Estimates on the log scale (excluding F6, which is in dollar levels) range from approximately -0.40 to $+0.23$.

The central pattern is that specifications without week fixed effects (E2 and E4) cluster in the negative tail of the distribution, while specifications with week fixed effects (E1 and E3) are more likely to produce positive or near-zero estimates. This is consistent with the concern that, in the absence of week FE, the staggered treatment rollout confounds the treatment effect with aggregate platform trends.

5.2 Meta-Regression

Table 3 reports the meta-regression results from Equation (2). The overall R^2 is 0.069, indicating that the four specification dimensions explain approximately 7% of the cross-specification variation in $\hat{\delta}$. While this is low in absolute terms, the partial R^2 decomposition reveals a clear ranking of dimension importance.

The fixed effects structure accounts for 61.8% of the explained variation. The coefficient on E2 (seller FE only, dropping week FE) is -8.83 (SE $= 4.42$, $p = 0.046$), indicating a large and statistically significant downward shift when week fixed effects are removed. This confirms that absorbing aggregate time trends is critical for identification in this staggered design: without week FE, the estimate captures the confounding effect of eBay-wide sales trends that differentially affected earlier versus later rollout cohorts.

The covariate design is the second most consequential dimension, accounting for 29.1% of

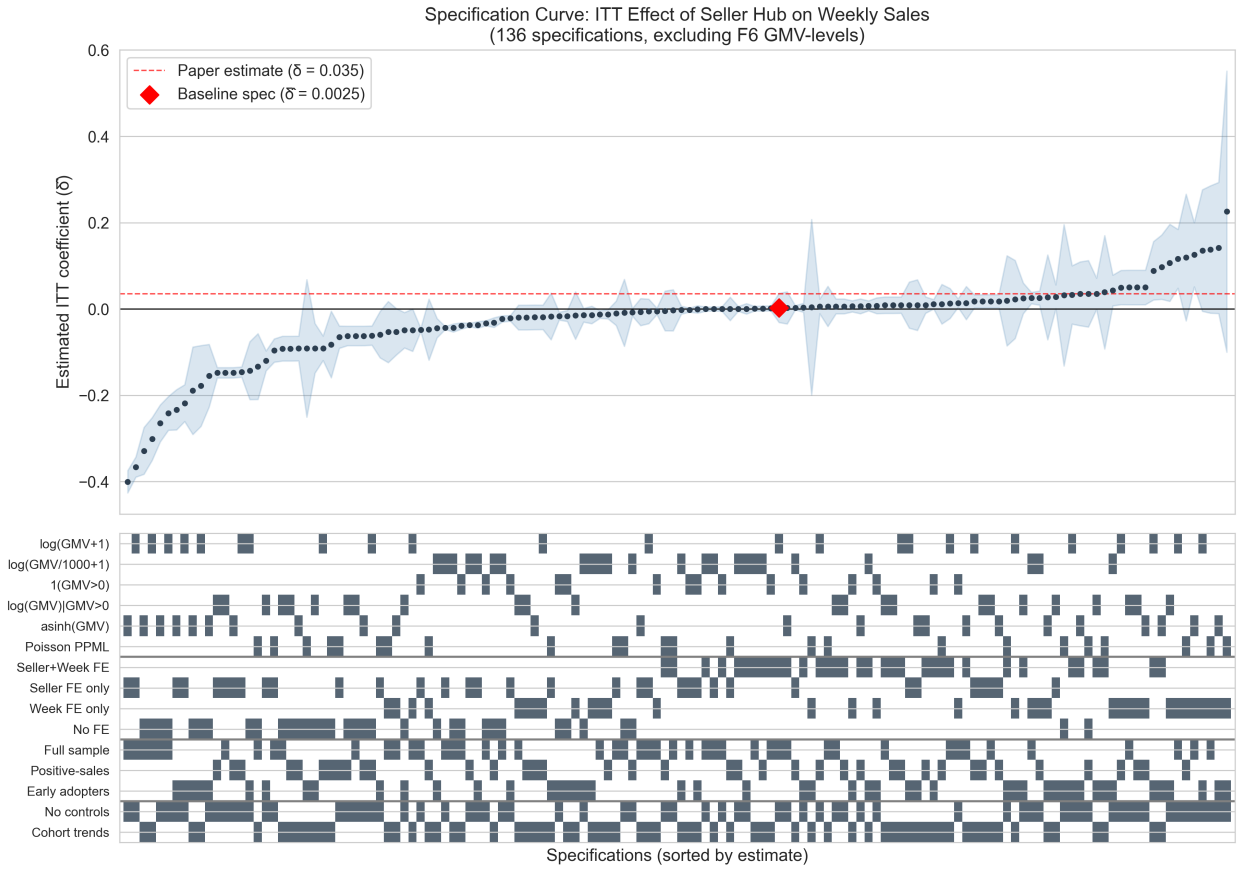

Figure 1: Specification Curve: 160 Specifications of the ITT Effect on Weekly Sales

Notes: Each point represents one specification. Estimates are sorted by magnitude. The lower panel indicates the functional form, fixed effects structure, sample, and covariate design for each specification. The dashed horizontal line marks zero. Results are from a 10% subsample; qualitative patterns are the focus. The original paper's baseline estimate is $\hat{\delta} = 0.035$ (Table 2, Col. 1).

explained variation. Adding cohort-specific linear time trends (D2) shifts the estimate by -9.98 (SE $= 5.59$, $p = 0.074$). This suggests that, even conditional on two-way fixed effects, allowing for differential trends across rollout cohorts further attenuates the estimated treatment effect.

Functional form contributes only 1.2% of explained variation. None of the functional form indicators are individually significant. This is a notable finding: despite the theoretical importance of the Chen and Roth (2024) critique, the choice among log-scale transformations, the extensive margin, Poisson PPML, and even raw dollar levels has little systematic effect on the estimate relative to the fixed effects structure. The large but imprecise coefficient on F6 (levels, -2.88, SE $= 18.15$) reflects that this specification operates on a different scale rather than producing a qualitatively different finding.

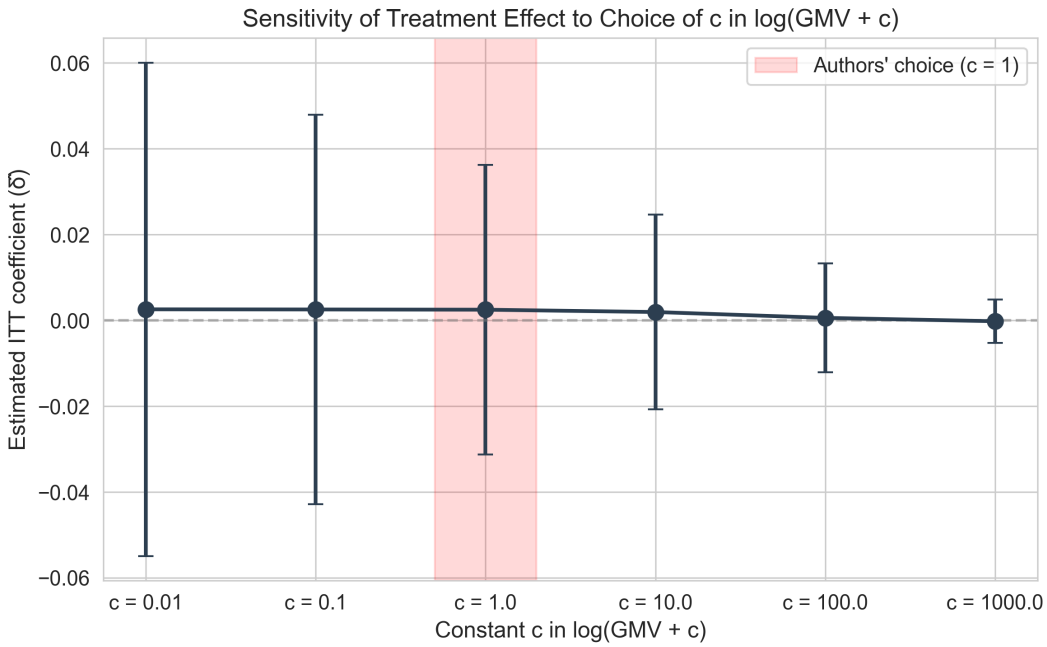

Figure 2: c-Sensitivity: Treatment Effect as a Function of the Additive Constant in $\log(\text{GMV}+c)$
Notes: The x-axis is the constant c in $\log(\text{GMV}+c)$ on a log scale. All specifications use baseline settings: seller+week FE, full sample, no additional controls. Dashed line marks zero. Confidence intervals use seller-clustered standard errors. Results from 10% subsample.

Sample definition accounts for 7.9% of explained variation. The early-adopter subsample (S3) produces larger positive estimates (+6.04, $p = 0.39$), consistent with complier heterogeneity reported in the original paper.

5.3 c-Sensitivity Analysis

Figure 2 shows the estimated treatment effect $\hat{\delta}$ as a function of the constant c in $\log(\text{GMV}+c)$, holding all other modeling choices at baseline. The estimate varies monotonically from $\hat{\delta} = 0.0026$ at $c = 0.01$ to $\hat{\delta} = -0.0002$ at $c = 1000$, crossing zero near $c = 100$. While all estimates are statistically insignificant in our 10% subsample, the qualitative pattern confirms the central argument of Chen and Roth (2024): the treatment effect estimate under $\log(1+y)$ depends on the arbitrary choice of the additive constant, and the original paper's use of $c = 1$ is one particular point along this continuum.

5.4 Supplementary: Quantity Outcome

As a robustness check, we re-estimate the baseline specification using $\log(\text{Quantity} + 1)$ as the dependent variable. The baseline ITT estimate is -0.0024 (SE $= 0.008$), close to zero and statistically insignificant. The extensive-margin effect (F3) is also near zero (0.00004, SE $= 0.003$). These patterns are consistent with the original paper's finding that SH affects revenue primarily through listing quality and pricing rather than through the number of transactions.

6 Conclusions

This paper conducts a specification curve analysis of the main ITT result in Bar-Gill, Brynjolfsson, and Hak (2024), systematically varying four dimensions of the baseline econometric model across 160 specifications. Our central finding is that the treatment effect estimate is stable within the space of identification-valid specifications, and that the dominant source of cross-specification variation is not functional form, the focus of much recent methodological debate, but the fixed effects structure that underwrites causal identification.

Specifications that include week fixed effects (E1 and E3) produce small estimates near zero, most of them positive, regardless of functional form or sample definition. Specifications that omit week fixed effects (E2 and E4) tend to produce negative estimates, often substantially so. The meta-regression confirms this ranking: the FE structure accounts for 61.8% of explained variation, while functional form contributes only 1.2%. This pattern is not evidence that the original result is fragile. It is evidence that the authors' specification choice was correct. In a staggered rollout where seven cohorts receive treatment at different calendar dates, week fixed effects absorb aggregate platform trends that would otherwise confound treatment timing with outcome trajectories (Goodman-Bacon, 2021). Removing them does not produce a "reasonable alternative" estimate; it removes the identification.

This distinction matters for how specification curve results should be interpreted. Not all 160 specifications in our analysis are equally credible. The framework of Simonsohn, Simmons, and

Nelson (2020) presents the distribution of estimates across modeling choices, but the researcher must still evaluate which specifications satisfy the identification requirements of the design. In this application, the set of identification-valid specifications, those that absorb common time shocks, yields a stable treatment effect near zero or modestly positive. The instability arises primarily from specifications that violate the identification strategy. This is informative about the design's dependence on time controls but not about the robustness of the causal conclusion.

Two subsidiary findings qualify this assessment. First, the c-sensitivity analysis confirms the Chen and Roth (2024) critique that $\log(1+y)$ estimates depend on the arbitrary additive constant. However, this variation is small in magnitude relative to the FE dimension, suggesting that functional form is a second-order concern in this application. Second, adding cohort-specific linear time trends (D2) attenuates the treatment effect ($\hat{\gamma}_D = -9.98$, $p = 0.074$), indicating that even within the identification-valid specification space, allowing for differential trends across rollout cohorts further reduces the estimated effect. This raises a question about whether the baseline two-way FE specification fully absorbs cohort-level confounding.

The broader implication is methodological. In applied program evaluation, researchers face many modeling choices, and specification curve analysis is a valuable tool for mapping out their consequences. The results of such an analysis are most informative when paired with a clear identification argument that distinguishes specifications that preserve causal identification from those that do not. In this application, the specification curve does not show that the treatment effect of data analytics on SME performance is ambiguous. It shows that correct specification of time fixed effects is the most important modeling decision in staggered experimental designs, and that the original authors made this decision correctly.

Table 3: Meta-Regression: Sources of Cross-Specification Variation

	Coefficient	Std. Error	p-value	Partial R^2 (%)
Intercept (γ_0)	4.430	4.786	0.355	
Functional form (ref: F1, $\log(GMV+1)$*)*				1.2
F2: $\log(\text{GMV}/1000+1)$	0.046	2.753	0.987	
F3: $\mathbf{1}(\text{GMV}>0)$	−0.373	3.709	0.920	
F4: $\log(\text{GMV}) \mid \text{GMV}>0$	0.032	2.750	0.991	
F5: $\text{asinh}(\text{GMV})$	−0.004	2.745	0.999	
F6: GMV levels	−2.879	18.145	0.874	
F7: Poisson PPML	0.053	2.747	0.985	
Fixed effects (ref: E1, Seller+Week FE)				61.8
E2: Seller FE only	−8.834	4.422	0.046	
E3: Week FE only	8.916	9.113	0.328	
E4: No FE	−8.463	5.462	0.121	
Sample (ref: S1, Full sample)				7.9
S2: Positive-sales only	1.764	5.974	0.768	
S3: Early adopters	6.041	7.009	0.389	
Covariate design (ref: D1, No controls)				29.1
D2: Cohort-specific trends	−9.982	5.589	0.074	
R^2	0.069			
F-statistic p-value	0.561			
N (specifications)	160			

Notes: OLS regression of specification-level point estimates $\hat{\delta}_k$ on indicators for each modeling choice (Equation 2). Omitted categories form the baseline specification: $\log(\text{GMV}+1)$, seller+week FE, full sample, no additional controls. Partial R^2 is computed as each dimension's sum of squares divided by the total model sum of squares. Results are from the 10% subsample.

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